👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.AT.11

Write Equations to Represent Relationships Between Two Variables

Lesson 9-3 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

Language Objective: I can define the variables in my equation and explain what each part represents using the words equation, variable, and rate.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

equation **variable** **rate** **represent** **define the variables** **Constant**

Level 3: try the blanks from memory first, then check the bank.

1. A mathematical sentence with an equals sign is an _____.

2. A letter standing for a quantity that can change is a _____.

3 A fixed amount of one quantity for every one unit of the other is a _____ .

4 To show a relationship with a table, graph, or equation is to _____ it.

5 To state what each letter stands for is to _____ _____ _____ .

6 A number in an equation that never changes is a _____ .

Write About the Math

Would you recommend that Diamond purchase or stream songs from the online music service?
¿Le recomendarías a Diamond comprar canciones o escucharlas en streaming del servicio de música en línea?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **equation** *ecuación*
- ☐ **variable** *variable*
- ☐ **rate** *tasa*
- ☐ **represent** *representar*
- ☐ **define the variables** *definir las variables*

Model: *The fee is charged EVERY month, so the total must be the rate multiplied by the number of months, not added to it once.* — A strong answer tests the equation against a table row (2 months should give $49.90) rather than just stating a rule. A weak answer picks multiplication without checking it against the actual numbers.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Diamond should write an ___ for each option so she can ___ the costs. Each equation uses the ___ as the input and gives the ___ as the output.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** equation
2. **Notes 2:** variable
3. **Notes 3:** rate
4. **Notes 4:** represent
5. **Notes 5:** define the variables
6. **Notes 6:** Constant

7. **Watch for this mistake:** *Writing the equation with addition — c = 24.95 + m — instead of multiplication. The fee is charged EVERY month, so the total is the rate times the number of months; always test the equation against a row of the table.*
8. **Try It 1:** D. The equation gives the total cost for ANY number of months, not just the rows listed — *A table can only show the values someone wrote down; substituting any value of m into c = 24.95m gives the cost for that many months instantly.*
9. **Try It 2:** A. 8 hours — *Divide both sides by 15: h = 120 ÷ 15 = 8.*